MSG No. 178.159 $P6_12'2'$ [Type III, hexagonal]

Table 1: Wyckoff site: `6a`, site symmetry: `.2'.`

No.	position	mapping
1	$[x,\, 0,\, 0]$	`[1,7]`
2	$\left[x,\, x,\, \frac{1}{6}\right]$	`[2,10]`
3	$\left[0,\, x,\, \frac{1}{3}\right]$	`[3,8]`
4	$\left[-x,\, 0,\, \frac{1}{2}\right]$	`[4,11]`
5	$\left[-x,\, -x,\, \frac{2}{3}\right]$	`[5,9]`
6	$\left[0,\, -x,\, \frac{5}{6}\right]$	`[6,12]`

Table 2: Wyckoff site: `6b`, site symmetry: `..2'`

No.	position	mapping
1	$\left[x,\, 2x,\, \frac{1}{4}\right]$	`[1,11]`
2	$\left[-x,\, x,\, \frac{5}{12}\right]$	`[2,9]`
3	$\left[-2x,\, -x,\, \frac{7}{12}\right]$	`[3,12]`
4	$\left[-x,\, -2x,\, \frac{3}{4}\right]$	`[4,7]`
5	$\left[x,\, -x,\, \frac{11}{12}\right]$	`[5,10]`
6	$\left[2x,\, x,\, \frac{1}{12}\right]$	`[6,8]`

Table 3: Wyckoff site: `12c`, site symmetry: `1`

No.	position	mapping
1	$[x,\, y,\, z]$	`[1]`
2	$\left[x-y,\, x,\, z+\frac{1}{6}\right]$	`[2]`
3	$\left[-y,\, x-y,\, z+\frac{1}{3}\right]$	`[3]`
4	$\left[-x,\, -y,\, z+\frac{1}{2}\right]$	`[4]`
5	$\left[-x+y,\, -x,\, z+\frac{2}{3}\right]$	`[5]`
6	$\left[y,\, -x+y,\, z+\frac{5}{6}\right]$	`[6]`
7	$[x-y,\, -y,\, -z]$	`[7]`
8	$\left[y,\, x,\, \frac{1}{3}-z\right]$	`[8]`
9	$\left[-x,\, -x+y,\, \frac{2}{3}-z\right]$	`[9]`
10	$\left[x,\, x-y,\, \frac{1}{6}-z\right]$	`[10]`
11	$\left[-x+y,\, y,\, \frac{1}{2}-z\right]$	`[11]`
12	$\left[-y,\, -x,\, \frac{5}{6}-z\right]$	`[12]`